Karra Gayatri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.43
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2022	98.70%
Blue Bells High School	Class X (Board of Secondary Education,Telangana)	2020	10.00%

SKILLS

Programming: Python, Java, C, C++, SQL

AI / Machine Learning:LLMs, RAG, Embeddings, TF-IDF, Logistic Regression, Linear Regression, Random Forest

Libraries & Frameworks: Streamlit, Sentence Transformers, FAISS, Pandas, NumPy, Scikit-learn, Matplotlib, PyMuPDF, Joblib

Tools: Git, GitHub, VS Code, MS Excel, MS Powerpoint

Leadership: Team Leadership, Project Coordination, Event Management, Communication, Problem Solving

PROJECTS

StudyOS -- Personal AI Learning Assistant

A RAG-based AI learning application for querying study material, generating quizzes, and analyzing previous question papers.

- **Technologies Used:** Python, Streamlit, PyMuPDF, Sentence Transformers, FAISS, RAG, LLM, OpenAI API, NumPy
- Built a **RAG pipeline** using **PDF extraction, text chunking, and embeddings** to retrieve relevant information across multiple study documents.
- Integrated an **LLM** with **FAISS vector search** to generate context-based answers with PDF and page references.
- Developed **AI-powered quiz generation and question-paper analysis** features to support exam revision and identify frequently tested topics.

Hybrid Email / Message Priority Classifier:

A machine learning application for classifying incoming emails and messages by priority and organizing them into a prioritized inbox.

- **Technologies Used:** Python, Pandas, Scikit-learn, TF-IDF, Logistic Regression, Streamlit, Joblib.
- Developed a **supervised text classification pipeline** utilizing **TF-IDF feature extraction** and **Logistic Regression** to categorize incoming messages across three distinct priority levels.
- Implemented **rule-based priority detection** with a **promotional safeguard**, preventing marketing messages containing urgency-related terminology from being incorrectly classified as High priority.
- Developed a **Streamlit-based interface** supporting **single-message classification** and **batch email analysis**, with concise **email summarization** and **priority-based inbox organization.**

Metro Demand Forecast:

A machine learning application for forecasting hourly metro demand and identifying the expected busiest period for a selected station and date.

- **Technologies Used:** Python, Pandas, Scikit-learn, Random Forest, SQLite, Streamlit, Matplotlib
- Processed **92,280 hourly ridership records across 83 stations**, engineering temporal and historical demand features to capture station-level usage patterns.
- Compared **Linear Regression and Random Forest** using chronological test data, with Random Forest achieving an **MAE of 34.26**.
- Developed a **Streamlit forecasting interface** that predicts hourly station demand and identifies the expected busiest period for a selected date.

RELEVANT COURSES

Introduction To Programming, Differential Equations, Linear Algebra, Complex Variables, Transform Techniques, Probability, Introduction to Statistics.

POSITIONS OF RESPONSIBILITY

National Service Scheme (NSS):

- **Head***(April 2026-Present)*
- Led campus-wide community service, rural development, and social welfare initiatives. Managed student volunteer teams and coordinated multiple outreach programs. Collaborated with faculty coordinators for planning and resource allocation.

58th Inter IIT Sports Meet & 30th Inter IIT Staff Sports Meet

- **Events Coordinator***(July 2025-Dec 2025)*
- Coordinated logistics, scheduling, and hospitality operations for a large-scale national sports event. Managed seamless execution across multiple venues and teams. Strengthened teamwork, communication, and event management skills.

EXTRACURRICULAR

- **Acting & Short Films:** Participated in multiple short-film projects and acting performances, developing communication, creativity, confidence, and teamwork skills.